Electrical Circuits II

(EE-1201)

Lecture-05

Network Theorems

(Maximum Power Transfer Theorem)

Dr. M. N. Alam, EED, NITW

Outlines

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This unit covers:

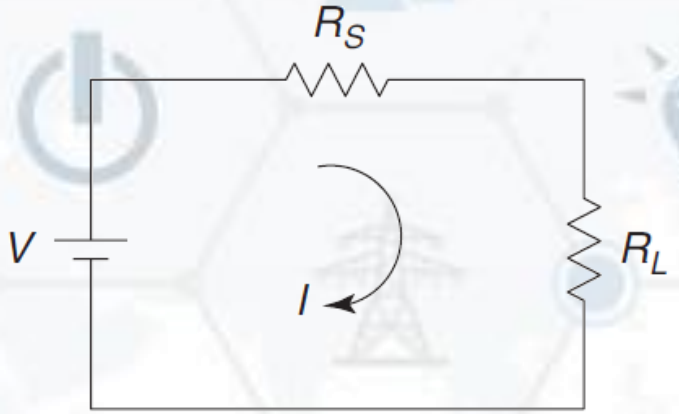
- Maximum power transfer theorem
- Numerical examples

Maximum Power Transfer Theorem

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- It states that *"the maximum power is delivered from a source to a load when the load resistance is equal to the source resistance."*

Proof:



- Current flowing in the circuit

$$I = \frac{V}{R_S + R_L}$$

- Power delivered to the load R_L

$$P = I^2 R_L = \frac{V^2}{(R_S + R_L)^2} R_L$$

- To determine the value of R_L for maximum power to be transferred to the load,

$$\frac{dP}{dR_L} = 0$$

$$\frac{dP}{dR_L} = \frac{d}{dR_L}\left[\frac{V^2}{(R_S + R_L)^2} R_L\right] = \frac{V^2[(R_S + R_L)^2 - (2R_L)(R_S + R_L)]}{(R_S + R_L)^4} = 0$$

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$$(R_S + R_L)^2 - 2R_L(R_S + R_L) = 0$$

$$R_S^2 + R_L^2 + 2R_S R_L - 2R_L R_S - 2R_L^2 = 0$$

$$R_S^2 - R_L^2 = 0$$

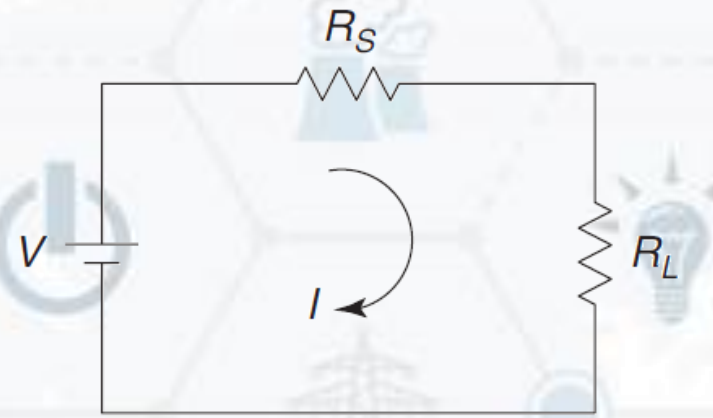
$$R_S = R_L$$

- Hence, the maximum power will be transferred to the load when load resistance is equal to the source resistance.

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During maximum power transfer:



- Current flowing in the circuit

$$I = \frac{V}{R_S + R_L} = \frac{V}{2R_S}$$

- Power delivered to the load R_L

$$P_{L,max} = I^2 R_L = \frac{V^2}{(2R_S)^2} R_S = \frac{V^2}{4R_S}$$

- Power supplied by the source

$$P_{Total} = I^2 R_L + I^2 R_S = \frac{V^2}{(2R_S)^2}(2R_S) = \frac{V^2}{2R_S}$$

- Efficiency during maximum power transfer,

$$\eta = \frac{Power\ delivered\ to\ the\ load}{Power\ supplied\ by\ the\ source} = \frac{V^2/4R_S}{V^2/2R_S} = \frac{1}{2} = 0.5\ (or\ 50\ \%)$$

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Steps to be Followed in Maximum Power Transfer Theorem

1. Remove the variable load resistance R_L
2. Find open circuit voltage V_{Th} across A and B
3. Find the resistance R_{Th} as seen from points A and B
4. Find resistance R_L for maximum power transfer

$$R_L = R_{Th}$$

5. Find the maximum power,

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{V_{Th}}{2R_{Th}}$$

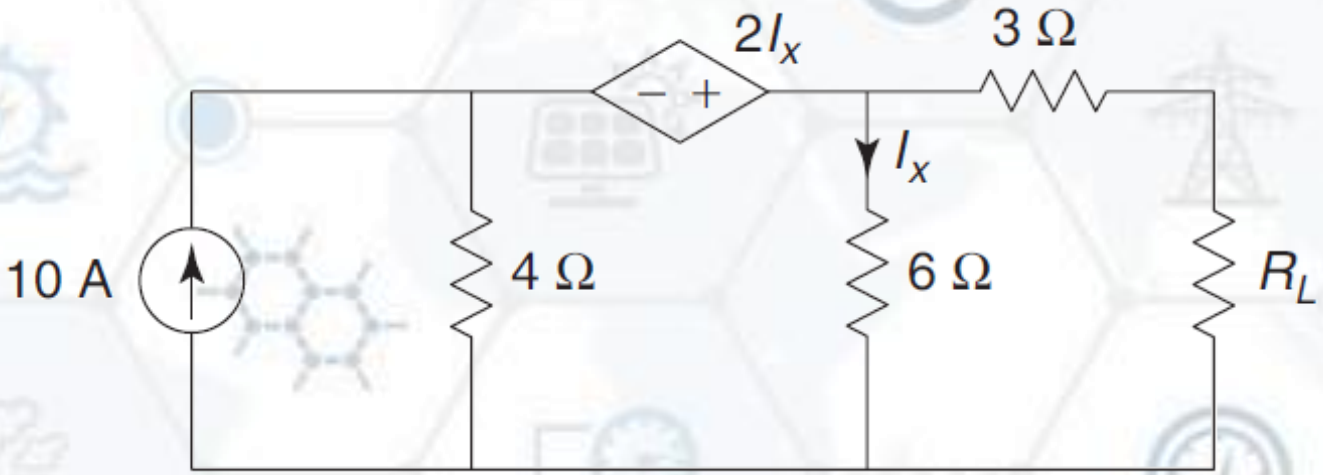
$$P_{max} = I_L^2 R_L = \frac{V_{Th}^2}{4R_{Th}^2} R_{Th} = \frac{V_{Th}^2}{4R_{Th}}$$

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Numerical Example-14:

- Find the value of resistance R_L for maximum power transfer and calculate the maximum power.

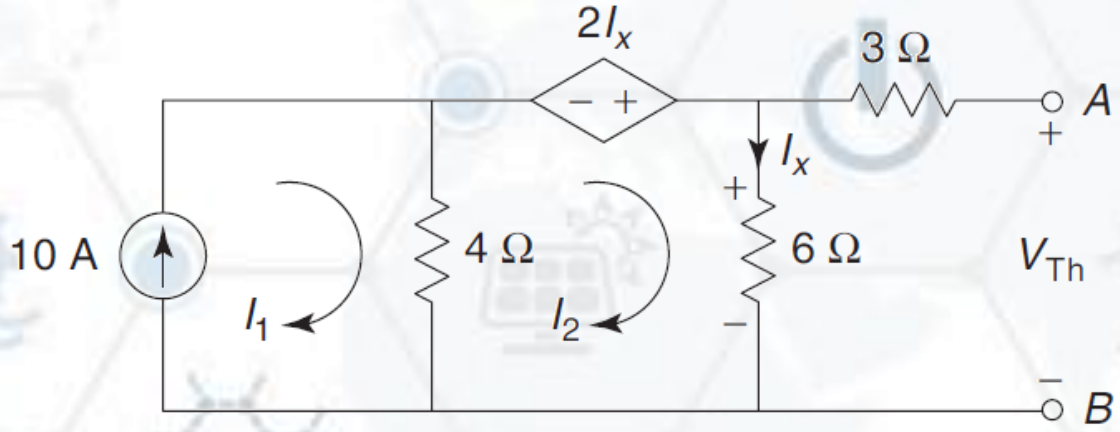


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Solution:

Step I: Calculation of V_{Th},



- From the figure, we can write that;

$$I_x = I_2 \ldots . (i)$$
$$I_1 = 10\ A \ldots . (ii)$$

- Applying KVL to Mesh 2,

$$2I_x - 6I_2 - 4(I_2 - I_1) = 0$$
$$2I_2 - 6I_2 - 4(I_2 - I_1) = 0$$
$$4I_1 - 8I_2 = 0 \ldots . (iii)$$

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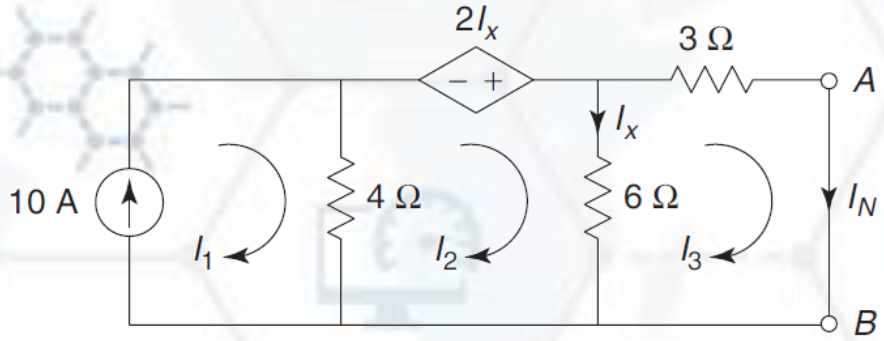
- Solving Eqs (ii) and (iii),

$$I_1 = 10\ A$$
$$I_2 = 5\ A$$

- Writing the V_{Th} equation,

$$6I_2 - 0 - V_{Th} = 0$$
$$V_{Th} = 6I_2$$
$$= 30\ V$$

Step II: Calculation of I_N,



- From the figure, we can write that;

$$I_x = I_2 - I_3 \ldots . (iv)$$
$$I_1 = 10\ A \ldots . (v)$$

Maximum Power Transfer Theorem

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- Applying KVL to Mesh 2,

$$2I_x - 6(I_2 - I_3) - 4(I_2 - I_1) = 0$$
$$2(I_2 - I_3) - 6(I_2 - I_3) - 4(I_2 - I_1) = 0$$
$$4I_1 - 8I_2 + 4I_3 = 0 \ldots . (vi)$$

- Applying KVL to Mesh 3,

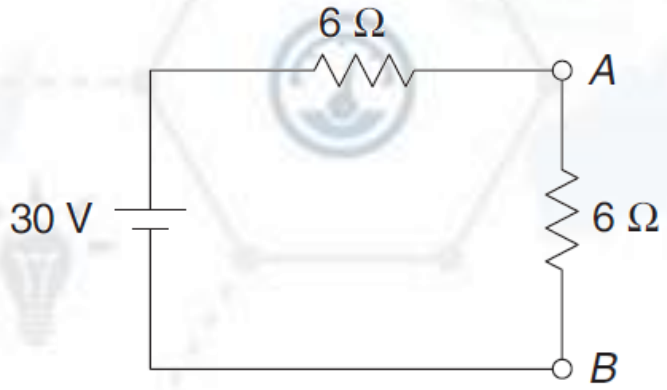
$$-3I_3 - 6(I_3 - I_2) = 0$$
$$6I_2 - 9I_3 = 0 \ldots . (vii)$$

- Solving Eqs (v), (vi) and (vii),

$$I_1 = 10\ A$$
$$I_2 = 7.5\ A$$
$$I_3 = 5\ A$$
$$I_N = 5\ A$$

Step III: Calculation of R_{Th},

$$R_{Th} = \frac{V_{Th}}{I_N} = \frac{30}{5} = 6\ \Omega$$



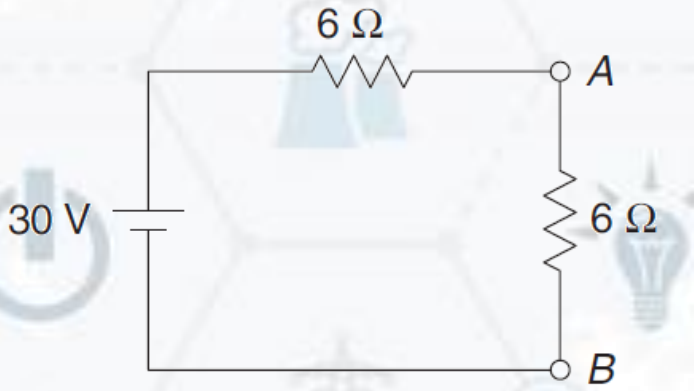
Maximum Power Transfer Theorem

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Step IV: Calculation of R_L,

- For maximum power transfer,

$$R_L = R_{Th} = 6\ \Omega$$



Step V: Calculation of P_{max},

- The maximum power transfer,

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

$$= \frac{30^2}{4 \times 6}$$

$$= 37.5\ W$$

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MATLAB Codes to Verify Maximum Power Transfer Theorem:

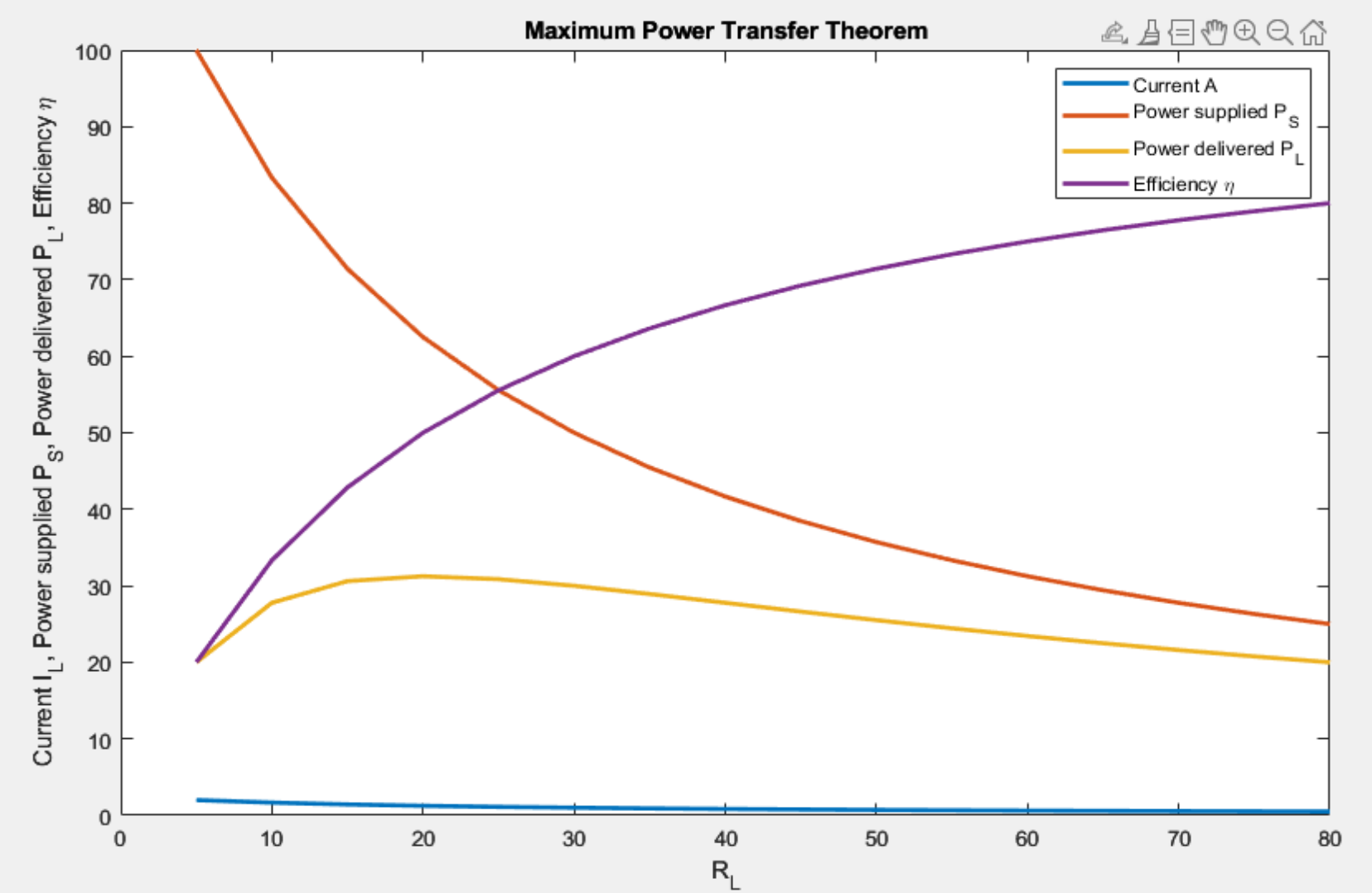
```
rs=20;
vs=50;
rl=5:5:80;

il=vs./(rs+rl);
ps=vs^2./(rs+rl);
pl=(vs^2./(rs+rl).^2).*rl;
ef=100*pl./ps;

plot(rl,il,rl,ps,rl,pl,rl,ef)
xlabel('R_L')
ylabel('Current I_L, Power supplied P_S, Power delivered P_L, Efficiency \eta')
legend('Current A', 'Power supplied P_S', 'Power delivered P_L' ,'Efficiency \eta')
title('Maximum Power Transfer Theorem')
```

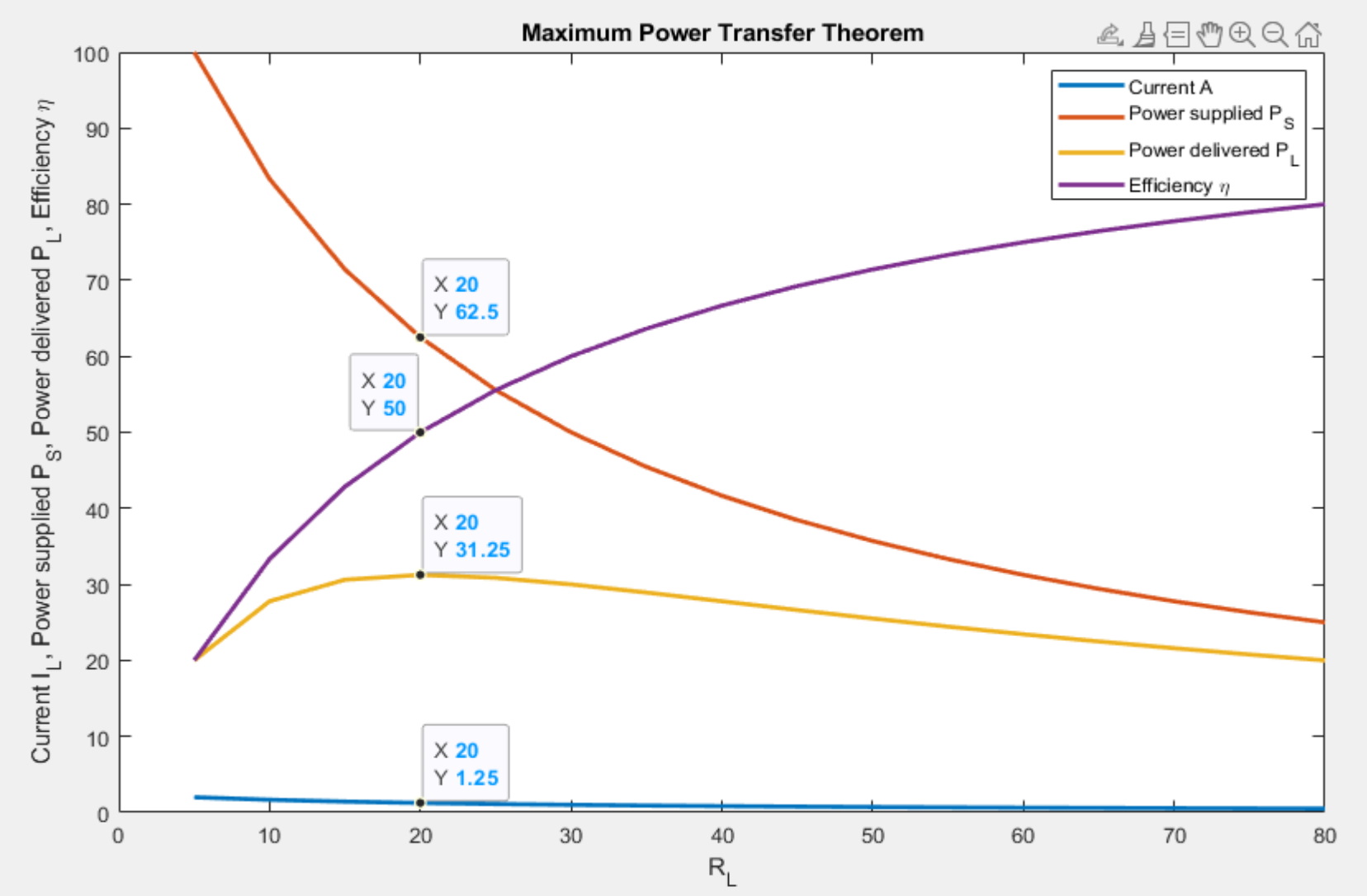
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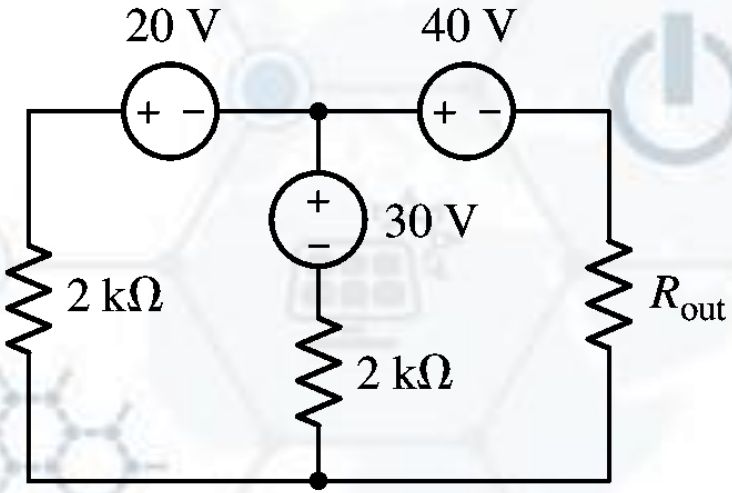


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Homework - #4

- In the circuit shown:



a) If $R_{out} = 3\ k\Omega$, find the power delivered to it.

b) What is the maximum power that can be delivered to any R_{out}?

c) What two different values of R_{out} will have exactly 20 mW delivered to them?

Ans: 230 mW; 306 mW; 59.2 $k\Omega$ and 16.88 Ω